Synoptic CB: Manual Redox Probes

2024 Samples

2026-06-17

Measurements taken with SWAP Redox Probes & Calomel Ref probe

Protocol: https://docs.google.com/document/d/

1dL2ZPhLFBAZJmVBCfKxup_KOYY96qA5IWfJmQen50pY/edit

Units = mV

##Add Required Packages

##Setup

```
###### File Names - PLEASE CHANGE
#file path and name for raw summary data file
dat <- read.csv("Raw Data/COMPASS_Synoptic_CB_Redox_2024_Raw.csv")
soildat <- read.csv("Raw Data/CB_Soil_pH.csv")

#file path and name of processed data file
processed_file_name = "Processed Data/COMPASS_Synoptic_CB_Redox_2024_processed.csv"

# ##Fix Sample Zone and Site naming
# soildat <- soildat %>%
#   mutate(Zone = case_when(Site == "SWH" & Zone == "Upland" ~ "Swamp",
#                           Site == "SWH" & Zone == "Upland-Control" ~ "Upland",
#                           TRUE ~ Zone),
#          Site = ifelse(Site == "Gcrew", "GCW", Site))

##Fix Site naming
dat <- dat %>%
  mutate(Site = ifelse(Site == "Gcrew", "GCW", Site))
dat$Site <- gsub(" ", "", dat$Site) ## Remove space in GWI site name

##Check redox depths of 30 cm
dat30 <- dat %>%
  filter(Depth == 30)

##Fix depths labelled incorrectly
```

```r
dat <- dat %>%
  mutate(Depth = case_when(Month == "October" & Site == "MSM" & Depth == 30 ~ 35,
                           Month == "October" & Site == "GWI" & Depth == 30 ~ 35,
                           Month == "November" & Site == "GWI" & Depth == 30 ~ 35,
                           Month == "November" & Site == "MSM" & Depth == 30 ~ 35,
                          TRUE ~ Depth))

##Fix sample dates labelled incorrectly as 2021
dat <- dat %>%
  mutate(Year = case_when(Month == "October" & Site == "SWH" & Year == 2021 ~ 2024,
                          TRUE ~ Year))


##Add column for SWH topography
dat <- dat %>%
  mutate(Topography = case_when(Site == "SWH" & Zone == "Transition-Hummock" ~ "Hummock",
                                Site == "SWH" & Zone == "Transition-Hollow" ~ "Hollow",
                                TRUE ~ NA),
         Zone = case_when(Site == "SWH" & Zone == "Transition-Hummock" ~ "Transition",
                          Site == "SWH" & Zone == "Transition-Hollow" ~ "Transition",
                          TRUE ~ Zone))

##Std error function
std.error <- function (x, na.rm=FALSE){
  sqrt(var(x, na.rm = na.rm) / sum(!is.na(x)))}
```

Adjust Raw Readings by pH

Need to adjust for pH:

redox = redoxi + [(ph - 5)59 + 224] This is the equation from Megonigal,

Patrick, & Faulkner 1993

They used a calomel electrode and used 224 to adjust to H2,

we use a KCl ref probe, so we need to add 202 instead

```r
##Remove unnecessary column in redox data
prelong <- dat %>%
  select (-Campaign) %>%
  rename (Notes = Notes.)

#switch redox data to long format
datlong <-  as.data.frame(melt(setDT(prelong), id.vars = c("Date", "Month", "Year", "Site", "Zone", "De
                               variable.name = "RP_mV"))
```

```r
# ##Add in soil pH data
# all_corr <- merge(datlong, soildat, by = c("Site", "Zone"), all = TRUE)
#
# ##Adjust redox for pH
# all_corr <- all_corr %>%
#   mutate(value_corr = value + ((pH - 5)*59 + 202))

##Adjust redox for KCl ref probe
datlong <- datlong %>%
  mutate(value_corr = value + 202)

##Average across all 5-6 measurements for each sampling Site, Zone, Depth, Month, Year
#Change all 30 cm measurements to 35 so depth is averaged from 30-35 cm
datlong1 <- datlong %>%
  mutate(Depth = case_when(Depth == 30 ~ 35,
                           TRUE ~ Depth))

red1 <- datlong1 %>%
  group_by(Date, Month, Year, Site, Zone, Depth, Time_24hr, Time_Zone, Topography, Notes) %>%
  summarise(Date = mean(Date),
            Avg_mV = mean(value_corr, na.rm=TRUE),
            Std_dv = sd(value_corr, na.rm=TRUE),
            Std_err = std.error(value_corr, na.rm=TRUE),
            Notes = first(Notes),
            Time_24hr = mean(Time_24hr),
            Time_Zone = first(Time_Zone)
            )

# head(red1)
```

##Write Out Adjusted & Avg'd Data

Vizualize Data